

STATISTICAL ANALYSIS AND HYPOTHESIS TESTING REPORT

1. DATA PREPROCESSING

The dataset was cleaned and standardized prior to analysis. Invalid age values (≤ 0 or > 100) were removed. Gender entries such as X, H, and - were normalized to "Unknown." Time was converted into hour format for analysis, and a binary feature was created for weapon usage. For stability in statistical testing, analysis was restricted to the top crime categories.

2. DESCRIPTIVE STATISTICAL ANALYSIS

2.1 Victim Age Analysis

The mean victim age is approximately 39.5 years, the median is 37, and the mode is 30. The standard deviation is approximately 15.57, and the range is 97. The distribution shows a slight positive skew (skewness $\approx +0.61$).

Interpretation:

The distribution is slightly right-skewed, indicating a longer tail toward older ages. Most victims fall within the young to middle-aged category. The relatively high standard deviation and wide range indicate that crime affects a broad age spectrum.

Insight:

Crime is not limited to a specific age group but is more concentrated among young and middle-aged individuals.

2.2 Gender Distribution

Male victims account for approximately 50.3%, female victims 48.2%, and unknown values 1.4%.

Interpretation:

The distribution is nearly balanced with a slight predominance of male victims. The low proportion of unknown values indicates good data quality.

Insight:

Crime affects both genders almost equally, with only a slight dominance of male victims.

2.3 Crime Severity Distribution

Medium severity crimes account for approximately 72%, while both severe and low severity crimes account for approximately 14% each.

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Interpretation:

Most crimes fall into the medium severity category, indicating that extreme and trivial crimes are less frequent.

Insight:

The majority of crimes are moderate in impact rather than extreme.

2.4 Premise Category Analysis

Residential areas account for approximately 39.5% of crimes, public/street locations 34.2%, and other categories form smaller proportions.

Interpretation:

Crimes are concentrated in both residential and public environments.

Insight:

Both private and public spaces act as major crime hotspots.

2.5 Crime Type Frequency

The most common crimes include battery (simple assault), burglary, theft, and vehicle-related crimes.

Interpretation:

There is a mix of violent crimes (assault) and property crimes (theft, burglary).

Insight:

The crime landscape consists of both violent and property-related offenses.

2.6 Time-Based Analysis

The mean occurrence time is around 1 PM, with a median of 2 PM. Crimes are distributed across all days of the week.

Interpretation:

Crimes tend to peak during afternoon hours and are evenly distributed throughout the week.

Insight:

Crime occurrence does not strongly depend on specific days but shows slight concentration during daytime hours.

2.7 Weapon Usage

Approximately 59% of crimes do not involve weapons, while 41% involve weapons.

Interpretation:

Although most crimes are non-weapon-based, a significant portion involves weapons.

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Insight:

There is a substantial presence of potentially violent incidents involving weapons.

2.8 Cross-Demographic Insight

Certain crime categories, particularly those involving children, show significantly lower average victim ages.

Insight:

Different crime types target different demographic groups.

2.9 Overall Statistical Summary

Descriptive statistical analysis reveals that crime is influenced by demographic, temporal, and environmental factors. Victim age shows a slight right-skew, with most victims in the young to middle-aged range. Gender distribution is nearly balanced. Crimes are predominantly of medium severity and occur mainly in residential and public areas. Crime occurrences are spread throughout the week, with slight concentration during afternoon hours. Weapon usage is present in a significant proportion of cases. Overall, crime patterns are structured and not random.

3. HYPOTHESIS TESTING

3.1 Hypothesis 1: Gender vs Crime Type

Null Hypothesis (H0): Gender and crime type are independent.

Alternative Hypothesis (H1): Gender and crime type are dependent.

Result:

p-value ≈ 0 , therefore H0 is rejected.

Interpretation:

Male victims are more prevalent in assault and burglary-related crimes. Female victims are more associated with intimate partner violence and identity theft.

Conclusion:

Victim gender is significantly associated with crime type.

3.2 Hypothesis 2: Age vs Crime Type

Null Hypothesis (H0): Mean victim age is the same across all crime types.

Alternative Hypothesis (H1): Mean victim age differs across crime types.

Result:

p-value ≈ 0 , therefore H0 is rejected.

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Interpretation:

Younger victims are more associated with intimate partner crimes, while older victims are more associated with burglary and property crimes.

Conclusion:

Victim age varies significantly across crime types.

3.3 Hypothesis 3: Weekend vs Crime Type

Null Hypothesis (H0): Crime type is independent of whether the incident occurred on a weekend.

Alternative Hypothesis (H1): Crime type depends on whether the incident occurred on a weekend.

Result:

p-value ≈ 0 , therefore H0 is rejected.

Interpretation:

Violent and interpersonal crimes are higher during weekends, while property crimes are more frequent on weekdays.

Conclusion:

Crime patterns differ between weekdays and weekends.

3.4 Hypothesis 4: Premise Category vs Crime Type

Null Hypothesis (H0): Crime type is independent of premise category.

Alternative Hypothesis (H1): Crime type depends on premise category.

Result:

p-value ≈ 0 , therefore H0 is rejected.

Interpretation:

Public areas show higher assault and vehicle-related crimes. Residential areas show higher domestic and identity-related crimes. Commercial areas are associated with theft and assault, while transit areas show a high concentration of vandalism.

Conclusion:

Crime type is strongly influenced by location.

4. FINAL COMBINED INSIGHT

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Hypothesis testing confirms that crime patterns are not random. Demographic factors (gender, age), temporal factors (weekend vs weekday), and environmental factors (premise category) all show statistically significant relationships with crime type. These variables serve as strong predictors and provide a foundation for machine learning models and further analysis.

5. NOTE ON STATISTICAL SIGNIFICANCE

Due to the large dataset size, p-values approach zero, indicating strong statistical significance. However, interpretation is based on distribution patterns and practical insights rather than p-values alone.